Demonstration of Proposed Features

Date	20 July 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Dataset EDA	Univariate, bivariate, multivariate analysis	Implemented	Yes	Shown via saved plots
2	Data Pre-processing Pipeline	Cleaning, encoding, scaling, train-test split	Implemented	Yes	
3	K-Means Clustering	Groups soil/climate profiles	Implemented	Yes	Exploratory insight, not the final classifier
4	Logistic Regression / KNN Classifier	Predicts the recommended crop	Implemented	Yes	Best model auto-selected and saved
5	Web Application (Flask)	Real-time crop prediction via browser form	Implemented	Yes	Live demo on localhost
6	Top-3 crop suggestions with confidence	Future enhancement, not yet built	Pending	No	Planned for future roadmap

Feature Implementation Summary

Metric	Value
Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83%